

1.3 Rewriting Expressions Using the Distributive Property

Lesson Introduction

 Benchmark	MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.		 Learning Targets	- I can use the distributive property to generate equivalent numeric expressions. - I can rewrite the sum of two composite numbers with a common factor using the distributive property.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.	
 Essential Questions	- What are the similarities and differences between writing equivalent expressions using the area model and the distributive property? - How are common factors used to generate equivalent expressions for a sum of two composite numbers?				
 Lesson Narrative	Students will extend prior knowledge of the distributive property to area models for rewriting numeric expressions. The lesson includes rewriting multiplication expressions and addition expressions using the distributive property and factors to build stronger foundations of equivalent numeric expressions before delving deeper into rewriting sums of composite numbers with a common factor. Students collaborate and discuss strategies to find missing values to make equations, comparing equivalent numeric expressions and analyzing errors when rewriting expressions using the distributive property. The Guided Instruction and Collaboration in this lesson provide opportunities to encourage flexible math thinking by rewriting equivalent expressions in various ways, including with common factors and the GCF.				
 Component Summary	1.3.1 Warm-Up (~5 min.)	Students use clues to identify a “secret number” (SE p. 15)	1.3.4 Guided Instruction (10 min.)	Analyze thinking and errors when generating equivalent expressions (SE p. 17)	
	1.3.2 Guided Instruction (10 min.)	Guiding prompts for applying the area model for representing distributive property (SE p. 15)	1.3.5 Collaboration (5-8 min.)	Work with a partner to find missing values in equations (SE p. 18)	
	1.3.3 Guided Practice (5 min.)	Practice rewriting expressions using the distributive property and area model (SE p. 16)	1.3.6 Your Turn! (5-7 min.)	Generate multiple equivalent expressions using common factors and the GCF (SE p. 19)	
	1.3.7 Wrap-Up (~5 min.)	Demonstrate understanding of learning targets by rewriting equivalent expressions using the distributive property (SE p. 19)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Distributive property of multiplication over addition - Greatest common factor (GCF)		(Optional) Chart paper (Optional) Sticky notes (Optional) Markers (Optional) Visual anchor chart for distributive property as area model		If using a shared space for students to contribute ideas during 1.3.6 Your Turn!, prepare chart paper or a whiteboard space with the expressions from questions 1a-d. Gather sticky notes or markers for students to write generated equivalent expressions.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students who struggle with multiplication may struggle to identify factors and equivalent expressions. - Students may confuse factors and multiples. - Students with gaps in conceptual understanding of area may have difficulty using the area model to show the distributive property. - Students may struggle to differentiate between expressions and equations.	- Consider using material and video from Lesson 1 to work on common factors. - While Guided Instruction and Your Turn include prompts to rewrite expressions using the GCF, not every expression needs to be rewritten using the GCF. Encourage flexible thinking to rewrite expressions using both the GCF and other common factors. - Throughout the lesson, math vocabulary is used to identify parts of an expression and answers. Consider reviewing the vocabulary words <i>sum</i>, <i>product</i>, <i>addend</i>, and <i>factor</i> before or during instruction. - If time is an issue, consider assigning 1.3.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, and Correct 1.3.4 Guided Instruction provides an opportunity for students to develop math reasoning by analyzing errors when generating equivalent expressions. Consider prompting students to identify incorrect use of common factors or the distributive property, then correcting errors.	MA.K12.MTR.4.1: Discussion and Reflection 1.3.4 Guided Instruction positions students to analyze the math thinking of others and lets students recognize errors to suggest how to correctly rewrite an equivalent expression. Consider using these opportunities to engage in discussions about mathematical ideas effectively using math vocabulary.
 Differentiate Instruction	1.3.5 Collaboration offers a variety of entry points for different learning styles and can offer a variety of ways for students to engage with the activity. Use informal observation to determine which method best fits your classroom.	
	1.3.6 Your Turn! - students can post generated expressions in a shared space (e.g., class whiteboard or chart paper) using sticky notes or markers. Use this alternative process to check understanding and engage in discussion on identifying equivalent expressions since some questions have more than one correct answer. For students who finish 1.3.6 or 1.3.7 early, consider extending the Secret Number activity as a challenge to complete with a friend or family member at home.	
Lesson Notes:		

1.3.1 Warm-Up: Secret Number

Component Overview: Students use a series of clues related to number properties to identify a secret number. Consider having students think of their own number, then use similar math vocabulary to create clues for somebody else to identify their number.



1.3.1 Warm-Up: Secret Number

Juanita has a secret number. Read her clues and answer the questions that follow.

Clue 1: My secret number is a factor of 60.

1. What is the smallest value Juanita's number could be?
1
2. What is the largest value Juanita's number could be?
60

Clue 2: My number is prime.

3. What values are possible for Juanita's number at this point?
2, 3, or 5

Clue 3: 15 is a multiple of my secret number.
3 or 5

Clue 4: My secret number is a factor of 20.

4. What is Juanita's secret number?
5
5. Could you have determined her secret number only using the first three clues? Explain.
No, it only narrowed the answer down to 2 possibilities



Looking for an engaging idea for a project? Consider extending this activity by allowing students to write a series of interesting clues using factors, multiples, and other properties needed for somebody else to identify their secret number. This could also be an engaging homework activity.

1.3.2 Guided Instruction: The Distributive Property

Component Overview: Students are introduced to the definition of the distributive property and extend their understanding to addition expressions. Instruction also focuses on flexibility to rewrite expressions using different addends within the parentheses. Students also practice using expressions written with the distributive property to find products using mental math.



1.3.2 Guided Instruction: The Distributive Property



The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property “distributes” a factor to all terms within a set of parentheses.

$$a \times (b + c) = (a \times b) + (a \times c)$$

The product 7×23 can be rewritten using the distributive property by decomposing 23 into two parts and multiplying 7 by both parts.

$$\begin{aligned} 7 \times 23 &= 7(20 + 3) \\ 7(20 + 3) &= (7 \cdot 20) + (7 \cdot 3) \end{aligned}$$

1. What are two other ways that 7×23 could be rewritten using the distributive property?

Answers will vary.

$$7 \times 23 = \underline{7} (\underline{10} + \underline{13})$$

$$7(23) = \underline{7} (\underline{11} + \underline{12})$$

2. Which of the expressions is the easiest to evaluate using mental math?

Answers will vary. $7(10 + 13)$; $7(20 + 3)$.

The distributive property can also be used to rewrite the sum of two numbers that share a common factor.



Consider building vocabulary and conceptual understanding of the distributive property by using the word “distribute” in a non-math context. Connect “distribute” to “give” with statements like, “If I distribute candy, that will mean **EVERY** person in this room would get a piece of candy. To distribute means to give.”



Have students practice mental math to find the product of an expression using the distributive property. For example, highlight that using the decomposed factors to multiply 7×20 and 7×3 , then adding the products, is easier than multiplying 7×23 mentally. Challenge students to generate another example and explain how mental math and distribution arrive at the same answer.

1.3.2 Guided Instruction: The Distributive Property (continued)

3. Elise rewrote the sum $27 + 18$ as $3(9 + 6)$. Are these two expressions equivalent?

- a. $27 + 18$ and $3(9 + 6)$ ☒ are ☐ are not equivalent

expressions because when evaluated they have

- ☒ the same value.
☐ different values.

- b. What common factor of 27 and 18 did Elise use to rewrite the expression?

3

- c. Rewrite the expression $27 + 18$ using a different common factor of 27 and 18. Use the area model to support your thinking.

	<u>3</u>	+	<u>2</u>
<u>9</u>	27		18

$$27 + 18 = \underline{9}(\underline{3} + \underline{2})$$



As an entry point for visual learners to “see” the distributive property, consider drawing arrows from the factor outside the parentheses to the addend inside the parentheses as you multiply to rewrite equivalent expressions as the sum of two products. Emphasize that the value is multiplied by both addends.



When rewriting sums, how can you determine which factor to distribute? Does the factor need to be the greatest common factor?

1.3.3 Guided Practice: Rewriting Expressions

Component Overview: Students synthesize learning of rewriting multiplication expressions in three ways: using the distributive property, modeling with area, and finding the product.



1.3.3 Guided Practice: Rewriting Expressions

- Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
98×5	$5(90 + 8)$	$ \begin{array}{r} 90 + 8 \\ 5 \begin{array}{ c c } \hline 450 & 40 \\ \hline \end{array} \end{array} $	490
6×22	<i>Answers will vary.</i> $6(12 + 10)$	$ \begin{array}{r} 12 + 10 \\ 6 \begin{array}{ c c } \hline 72 & 60 \\ \hline \end{array} \end{array} $	132

- Complete the table by rewriting each addition expression using the distributive property, modeling with area, and finding the sum.

Addition Expression	Distributive Property	Area Model	Sum
$600 + 140$	<i>Answers will vary.</i> $20(30 + 7)$	$ \begin{array}{r} 30 + 7 \\ 20 \begin{array}{ c c } \hline 600 & 140 \\ \hline \end{array} \end{array} $	740
$56 + 32$	<i>Answers will vary.</i> $8(7 + 4)$	$ \begin{array}{r} 7 + 4 \\ 8 \begin{array}{ c c } \hline 56 & 32 \\ \hline \end{array} \end{array} $	88



The second row of each table has multiple correct answers using various equivalent expressions. Encourage discourse on how various factors, not just the GCF, can be used to rewrite the expression using the distributive property. Use math vocabulary like “equal to” and “equivalent expressions” to highlight that the expressions are still equivalent.



When finding the product for these expressions, how can decomposing factors make it easier to multiply using mental math?



Consider an error analysis for the distributive property expressions in the question 2 table. For example, look at student work to identify a “My Favorite No.” Use this opportunity to emphasize procedural knowledge and a growth mindset by learning from mistakes.

1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property

Component Overview: Students rewrite the sum of 16 and 24 using common factors and the distributive property, then analyze errors in rewritten expressions that are not equivalent. This activity provides a further opportunity for group or partner discussion to recognize, correct, and clarify procedural use of the distributive property to rewrite equivalent expressions.



1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property

Sums of **composite whole numbers** can be rewritten using the distributive property and a common factor of the numbers.



A **composite number** is a whole number greater than 1 that has at least one whole-number factor other than one and itself.

- What are the common factors of 16 and 24?

Factors of 16	1, 2, 4, 8, 16
Factors of 24	1, 2, 3, 4, 6, 8, 12, 24
Common Factors of 16 and 24	1, 2, 4, 8

- Isaiah rewrote the sum $16 + 24$ using the distributive property. Each step in his thinking is shown. Complete the descriptions of Isaiah's process by examining his thinking and work. Use the words in the bank below to complete each description.

distributive equivalent factor multiplication

Isaiah's Thinking (💭) and Work (✍️)	Description
💭 "I know 2 is a common factor of both 16 and 24."	Isaiah identified a <u>factor</u> both numbers have in common.
✍️ $16 + 24 = 2(8) + 2(12)$	He rewrote the expression using <u>multiplication</u> with the common factor.
✍️ $2(8) + 2(12) = 2(8 + 12)$	He rewrote the expression using the <u>distributive</u> property.
💭 "The sum $16 + 24$ can be rewritten as $2(8 + 12)$ using the distributive property."	Isaiah recognized that $16 + 24$ and $2(8 + 12)$ are <u>equivalent</u> numeric expressions.



Consider activating prior knowledge from previous lessons:

- What is a factor? What is a common factor?
- Why would we need to be looking for common factors?
- Which factors can be used to create equivalent expressions?



Notice the icons used to represent different parts of the working process. Consider using these "thinking" and "work" prompts for your students to create additional opportunities for them to make sense of information.

1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property (continued)

3. Rewrite the sum $16 + 24$ using the distributive property in two additional ways.

$$16 + 24 = 8(2 + 3)$$

$$16 + 24 = 4(4 + 6)$$

$2(8 + 12)$ is also a possible solution.

4. Two students made errors rewriting the sum $54 + 36$ using the distributive property. Each student's expression is shown.

Rewrite $54 + 36$ using the distributive property.	
Naveah's Expression	$4(14 + 9)$
Gavin's Expression	$3(18 + 36)$

Work with your partner to answer the following questions.

- What error did Naveah make?
 $4 \cdot 14 \neq 54$
 4 is not a common factor of 54 and 36 .
 - What error did Gavin make?
 $Gavin$ forgot to factor the 3 out of the 36 .
 - Generate two different equivalent expressions for the sum $54 + 36$ by rewriting it using the distributive property.
Answers will vary.
 $2(27 + 18)$
 $3(18 + 12)$
5. Rewrite an equivalent expression for the sum of $40 + 64$ using the distributive property with the **greatest common factor** (GCF).
 $8(5 + 8)$



The error analysis embedded within this activity gives students additional practice on MA.K12.MTR.4.1 and higher order thinking.



Clarify, Critique, and Correct

Consider emphasizing and asking students to identify and share reasons for why the errors lead to expressions that are not equivalent. Focus on reasoning connected to the correct use of the distributive property and identification of common factors.

1.3.5 Collaboration: Missing Values

Component Overview: Students work in partnerships to write missing values that make an equation true and write about how they determined the missing value. Consider having students verbalize reasoning for the values they used to make the equations true.



1.3.5 Collaboration: Missing Values

- Work with your partner to determine the missing values that will make each equation true.

a. $40 + 16 = 4(10 + 4)$ b. $15 + 80 = 5(3 + 16)$

c. $108 + 198 = 18(6 + 11)$ d. $28 + 63 = 7(4 + 9)$

e. $45 + 18 = 3(15 + 6)$ f. $135 + 27 = 9(15 + 3)$

- Choose one of the equations from question 1 and explain how you determined the missing value(s).
Answers will vary.

To find the missing value in equation _____, I
_____.

- Have your partner read your explanation and offer suggestions on how to improve it.



Determine how students are collaborating based on informal observations from the first few activities.

- Have students split the work (1a-c and 1d-f) and share their answers and reasoning.
- Have students work together, with one student only allowed to talk (progressing student) and other taking notes (proficient student).
- Have students choose which three to engage with (increasing choice and entry point) and find a partner around the room who did the same three to compare answers for questions 2 and 3.



Notice additional entry points using sentence stems in question 2 and intentional sharing for auditory learners in question 3. Consider using vocabulary cards to encourage students to replace words like “same number” with “common factor” when articulating their responses.

1.3.6 Your Turn!

Component Overview: Students work independently to rewrite expressions using common factors, including the GCF and the distributive property. Consider using this practice to encourage collaboration, flexible thinking, and discussion by allowing students to post generated equivalent expressions in a shared space.



1.3.6 Your Turn!

1. Rewrite an equivalent expression for each sum using the distributive property with a common factor.

a. $51 + 21$
 $3(17 + 7)$

b. $124 + 20$
 $2(62 + 10)$ or $4(31 + 5)$

c. $72 + 24$
 $24(3 + 1)$
 $12(6 + 4)$
 $8(9 + 3)$
 $6(12 + 4)$
 $4(18 + 6)$
 $3(24 + 8)$
 $2(36 + 12)$

d. $39 + 65$
 $13(3 + 5)$

2. Rewrite an equivalent expression for the sum of $68 + 34$ using the distributive property with the **greatest common factor** (GCF).
 $34(2 + 1)$



Challenge students who finish early to create as many equivalent expressions as possible. They will notice that 1a and 1d only have one common factor, while option 1c has a variety.

1.3.7 Wrap-Up: Ticket Out the Door

Component Overview: Students apply their understanding to rewrite sums of two composite numbers using common factors and the distributive property.



1.3.7 Wrap-Up: Ticket Out the Door

1. Rewrite an equivalent expression for the sum of $75 + 30$ in two different ways using the distributive property and a common factor.

Answers will vary.

Possible answers include:

$15(5 + 2)$ or $5(15 + 6)$ or $3(25 + 10)$

2. Complete the equation with the missing values that make the equation true.

$$49 + 35 = 7 (7 + 5)$$



Consider supplementing these questions by asking students to answer one of the essential questions for this lesson in their own words. The essential questions are open-ended and allow students to represent their thinking in a format that best communicates their understanding. For example, one student could create a visual, step-by-step guide with labels explaining their response.